CHAPTER 03

Bonding

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: f7c53a054a3e64cf4906eea887de66adcae1abb76ae0433ec58f68af5fd4c127

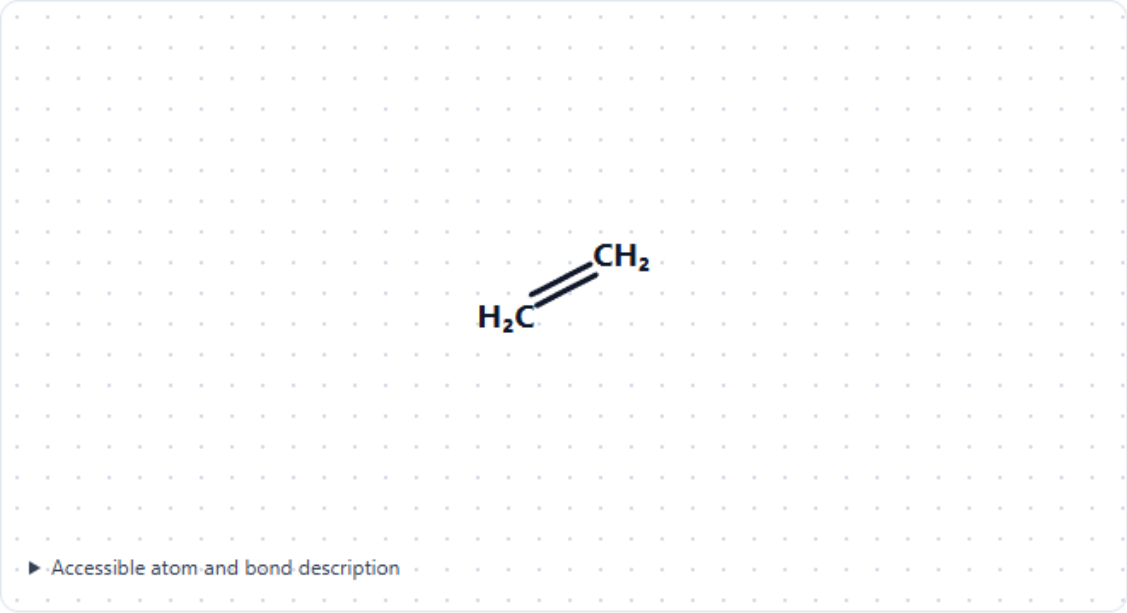
BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

quantum | Independent exercise

The ethene drawing is a companion bonding example only. These questions concern atomic orbitals and subshells, not the total electrons or molecular orbitals of C2H4. No atomic filling diagram is supplied during this workshop.

Screenshot: quantum-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2]]

quantum:stage-1 | For a 2p electron, which n and l pair is allowed?

Answer type: choice

Option 0 (A): n=2, l=1

Option 1 (B): n=2, l=2

Option 2 (C): n=1, l=1

quantum:stage-2 | How many orbitals are in a p subshell?

Answer type: number | unit: orbitals

quantum:stage-3 | How many electrons can a filled p subshell hold?

Answer type: number | unit: electrons

quantum:stage-4 | For a hydrogen-like 3p orbital, how many radial nodes are present?

Answer type: number | unit: radial nodes

quantum:stage-5 | How many angular nodes does that 3p orbital have?

Answer type: number | unit: angular nodes

quantum:stage-6 | In the usual multielectron-atom model, which listed n=3 subshell is lowest in energy?

Answer type: choice

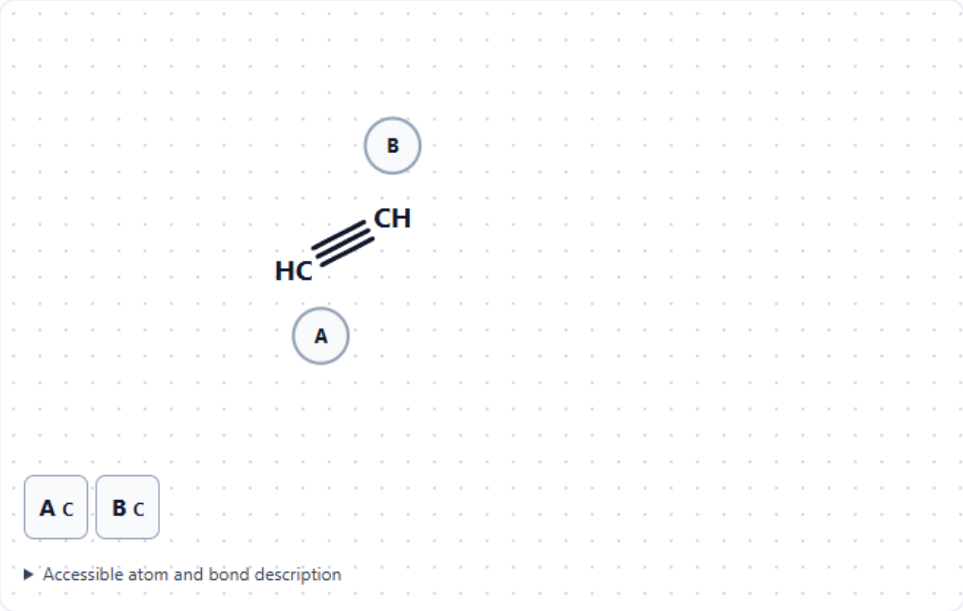
Option 0 (A): 3s

Option 1 (B): 3p

Option 2 (C): 3d

bond-count | Independent exercise

Screenshot: bond-count-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 3]]

bond-count:stage-1 | Select the atoms joined by the triple bond.

Answer type: atoms

bond-count:stage-2 | How many pi bonds are present?

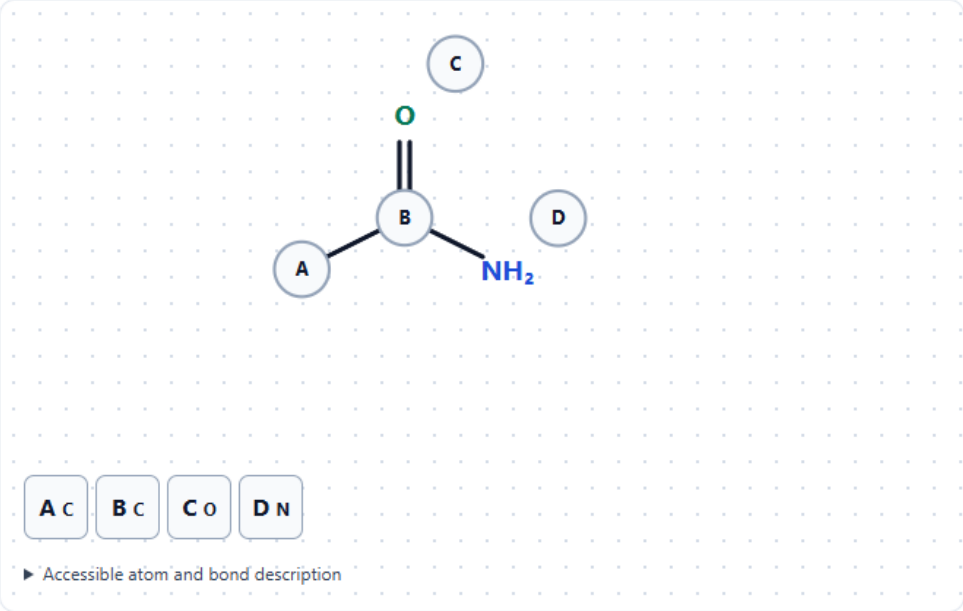
Answer type: number | unit: pi bonds

bond-count:stage-3 | Including the C–H bonds represented by the CH labels, how many sigma bonds are present?

Answer type: number | unit: sigma bonds

amide-resonance | Independent exercise

Screenshot: amide-resonance-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

amide-resonance:stage-1 | Select the amide nitrogen.

Answer type: atoms

amide-resonance:stage-2 | Why is the amide group approximately planar?

Answer type: choice

Option 0 (A): Nitrogen lone-pair conjugation with C=O

Option 1 (B): All nitrogen atoms are planar

Option 2 (C): C–N is a freely rotating pure single bond

amide-resonance:stage-3 | For acetamide, resonance contributors differ in…

Answer type: choice

Option 0 (A): Positions of atoms

Option 1 (B): Electron placement only

Option 2 (C): Molecular formula